

```

# Encoding of score of subject in a qubit
for i, sscore in enumerate(sscores):
    qml.RX(sscore, swires=i)
# Variational layer
for i in range(3):
    qml.RY(sweights[i], wires=i)
return qml.expval(qml.PauliZ(0))

# Student Data
sscores = np.array([0.7, 0.6, 0.9])
sweights = np.array([0.2, 0.4, 0.7], requires_grad=True)
sprediction = studentcircuit(sscores, sweights)
print("Prediction:", "Pass" if sprediction > 0 else "Fail")

```

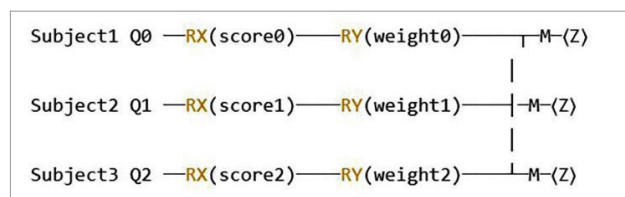


Figure 3: Quantum circuit for student assessments

This QML-based approach helps assess teaching and learning processes with a higher degree of performance and accuracy.

A few key use cases of QML-based quantum computing for academics are:

- Optimisation of grading rubrics
- Weighing of questions
- Personalised feedbacks
- Hidden patterns analysis in attendance, grades and participation
- Performance predictions

There is huge scope for research in quantum data science, quantum machine learning and quantum deep learning. Traditional implementations of classification, regression, and clustering can be improved using quantum circuits. Quantum computing enables multi-dimensional analytics in minimum time with maximum accuracy. Researchers can develop and simulate novel quantum circuits for solving real world problems including portfolio optimisation in investments (banking and finance), prediction of protein folding (disease research), quantum key distribution (secured communications), traffic management (smart cities), and climate modelling (disaster prediction). **END** 

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